

TECHNICAL REPORT

Basic Linux Administration and Backup Automation

Ubuntu 26.04 LTS on Oracle VirtualBox

Field	Details
Author	Oscar
Date	May 2026
Environment	Oracle VirtualBox
Operating system	Ubuntu 26.04 LTS
Areas covered	Users and groups, permissions, SGID, Bash scripting, and cron

Technical note. The original evidence corresponds to an Ubuntu 26.04 LTS installation with a graphical environment. For accuracy, this document uses the term “Ubuntu 26.04 LTS” instead of “Ubuntu Server.”

1. Overview

This report documents the initial configuration of an Ubuntu 26.04 LTS environment virtualized with Oracle VirtualBox and the execution of fundamental Linux system administration tasks.

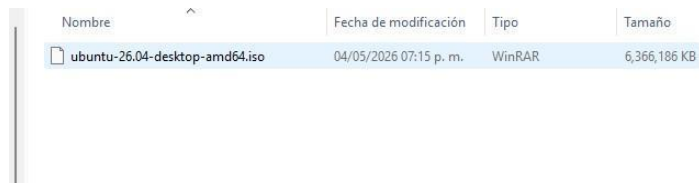
The activities performed include:

- Linux environment setup and system updates;
- basic user and group administration;
- creation of shared directories and permission configuration;
- use of the SGID bit for group inheritance;
- creation of a Bash script to back up the /home directory;
- backup automation using cron;
- verification of the configurations and the generated backup file.

2. Environment Preparation and Installation

2.1 Installation Image Used

An official Ubuntu 26.04 LTS Desktop ISO image was used as the base for the virtual machine.

A screenshot of a file manager window showing a single file. The file is named 'ubuntu-26.04-desktop-amd64.iso', has a size of 6,366,186 KB, and was modified on 04/05/2026 at 07:15 p. m. The file type is listed as WinRAR.

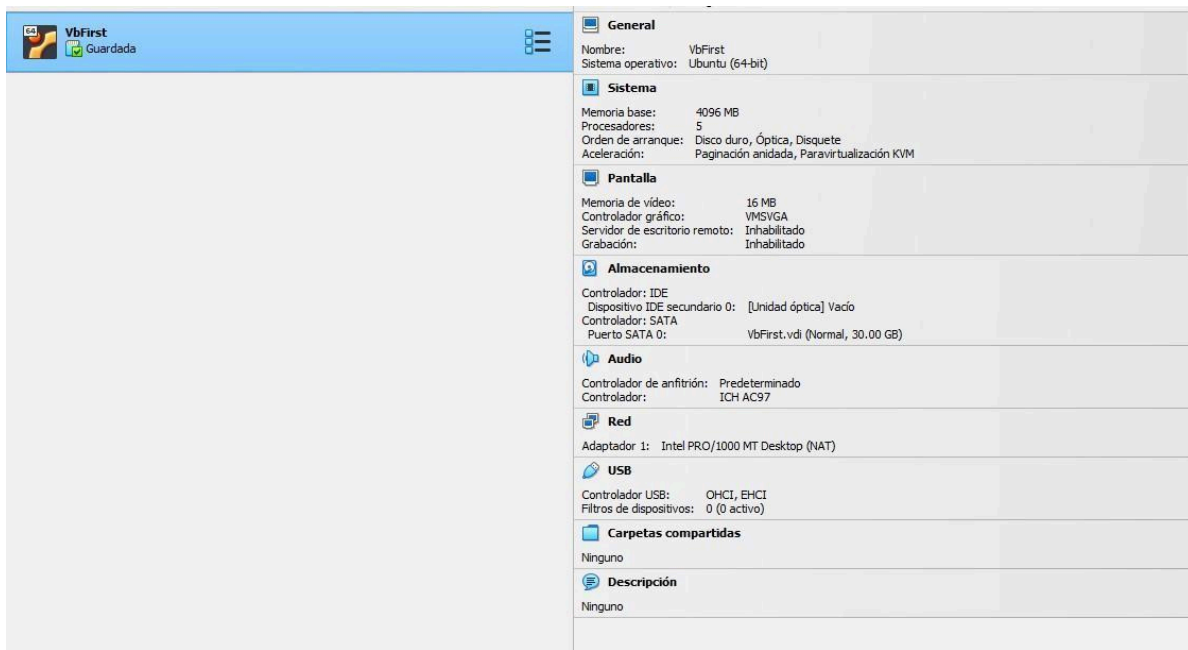
Nombre	Fecha de modificación	Tipo	Tamaño
ubuntu-26.04-desktop-amd64.iso	04/05/2026 07:15 p. m.	WinRAR	6,366,186 KB

Evidence 1. ISO file used for the installation.

2.2 Virtual Machine Creation

The virtual machine was created in Oracle VirtualBox with sufficient resources to perform the administration and automation tasks included in the exercise.

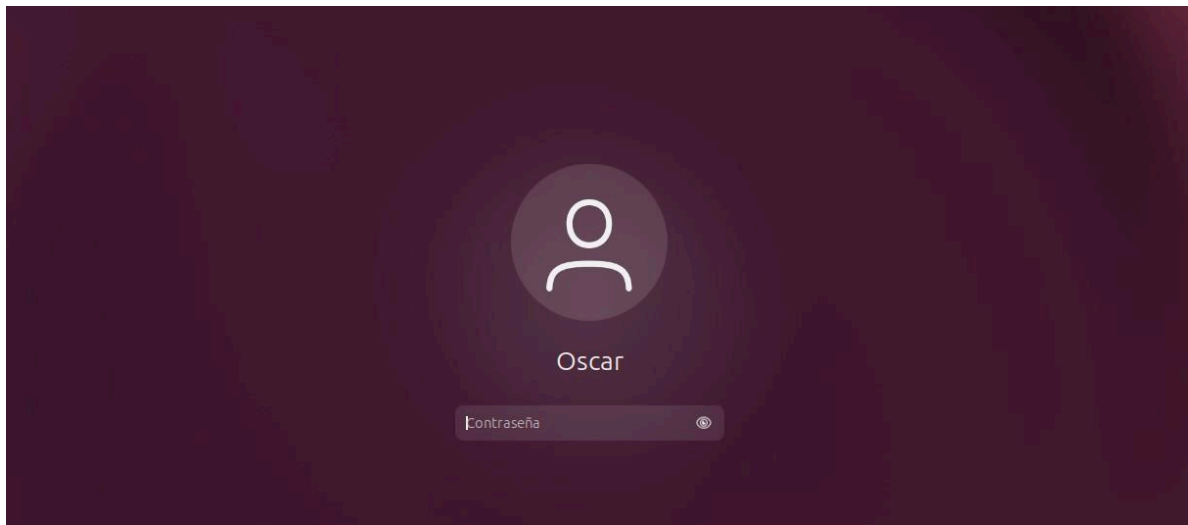
Resource	Configuration
CPU	5 cores
RAM	4 GiB
Virtual disk	30 GiB
System type	Ubuntu (64-bit)
Virtualization	Oracle VirtualBox



Evidence 2. Virtual machine configuration summary.

2.3 System Installation and Access

The operating system was installed using the Ubuntu installer. The virtual machine was identified as “VbFirst” and the user “Oscar” was configured. The password used is not documented in this report.



Evidence 3. Ubuntu environment login screen.



Evidence 4. Terminal session successfully started.

2.4 System Update

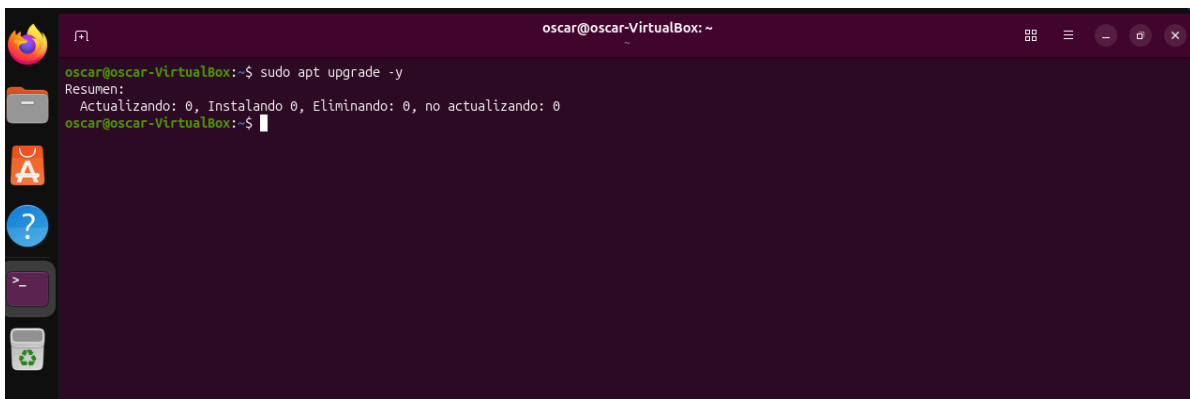
After logging in, the package lists were updated, and the available package upgrades were installed.

```
sudo apt update && sudo apt upgrade -y
```

The Linux distribution and kernel were verified from the terminal. A subsequent update check showed that no packages were pending at that time.

```
oscar@oscar-VirtualBox:~$ uname -a
Linux oscar-VirtualBox 7.0.0-15-generic #15-Ubuntu SMP PREEMPT_DYNAMIC Wed Apr 22 16:06:43 UTC 2026 x86_64 GNU/Linux
oscar@oscar-VirtualBox:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:   Ubuntu 26.04 LTS
Release:      26.04
Codename:     noble
```

Evidence 5. Distribution and kernel information.



Evidence 6. Package update status verification.

3. User and Group Management

3.1 User Creation

Test users with different functional roles were created to practice account organization and permission management.

User	Role used in the exercise
jdoe	Administration
asmith	Development
bwhite	Development

The “Administration” label describes the role used in the exercise; membership in the custom admin group does not by itself grant sudo privileges.

3.2 Group Creation

Two functional groups were created:

```
sudo groupadd admin
sudo groupadd desarrolladores
```

3.3 Assigning Users to Groups

```
sudo usermod -aG admin jdoe
sudo usermod -aG desarrolladores asmith
sudo usermod -aG desarrolladores bwhite
```

3.4 Membership Verification

Group memberships were verified with:

```
groups jdoe
groups asmith
groups bwhite
```

The configuration was also verified by inspecting the `/etc/group` file to confirm that the users were associated with the corresponding groups.

```
gdm:x:177:
jdoe:x:1001:
asmith:x:1002:
bwhite:x:1003:
admin:x:1004:jdoe
desarrolladores:x:1005:asmith,bwhite
```

Evidence 7. Group and membership verification.

4. Shared Directories and Permissions

4.1 Directory Creation

```
sudo mkdir /home/proyectos
sudo mkdir /home/admin
```

Directory	Purpose
/home/proyectos	Shared files for the desarrolladores group
/home/admin	Shared files for the admin group

4.2 Project Directory Permissions

```
sudo chown :desarrolladores /home/proyectos
sudo chmod 770 /home/proyectos
```

Mode 770 grants read, write, and execute permissions to the owner and group while granting no permissions to other users.

4.3 Administrative Directory Permissions

```
sudo chown :admin /home/admin
sudo chmod 770 /home/admin
```

4.4 Automatic Group Inheritance with SGID

The SGID bit was enabled on both directories so that new files and subdirectories inherit the group of the parent directory.

```
sudo chmod g+s /home/proyectos
sudo chmod g+s /home/admin
```

After applying SGID, the group execute position appears as s (for example, drwxrws---), confirming that the bit is enabled.

```
oscar@oscar-VirtualBox:~$ ls -ld /home/proyectos
drwxrws--- 2 root desarrolladores 4096 May  5 20:29 /home/proyectos
oscar@oscar-VirtualBox:~$ ls -ld /home/admin
drwxrws--- 2 root admin 4096 May  5 20:30 /home/admin
oscar@oscar-VirtualBox:~$
```

Evidence 8. Permissions for /home/proyectos and /home/admin with SGID enabled.

5. Backup Script and Automation

5.1 Script Creation

The following file was created:

```
/usr/local/bin/respaldo_home.sh
```

The purpose of the script is to generate a compressed backup of the contents of /home inside /var/backups.

5.2 Script Content

The script is shown as originally implemented. Its filename, comments, and output paths are preserved to remain consistent with the project evidence.

```
#!/bin/bash
# Script de respaldo automático en Linux
#
# Crea un archivo comprimido (.tar.gz) con el contenido de /home.
# El nombre del respaldo incorpora la fecha de ejecución.
tar -czf /var/backups/respaldo_home_$(date +%F).tar.gz /home
```

The tar command uses the options -c (create archive), -z (compress with gzip), and -f (specify the output file). The expression \$(date +%F) adds the date in YYYY-MM-DD format.

Technical note. With the intended once-daily schedule, including the date prevents backups from previous days from being overwritten. If the script were run multiple times on the same day, it would use the same filename and could replace that day's backup.

5.3 Execution Permissions

```
sudo chmod +x /usr/local/bin/respaldo_home.sh
```

5.4 Automatic Scheduling with cron

The task was configured in root's crontab to run the backup every day at 02:00.

```
sudo crontab -e
0 2 * * * /usr/local/bin/respaldo_home.sh
```

Using root's crontab is consistent with the backup location in /var/backups and facilitates access to the different user directories included under /home.

```

GNU nano 8.7.1 /tmp/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
0 2 * * * /usr/local/bin/respaldo_home.sh

```

Evidence 9. Crontab entry scheduled to run the backup daily at 02:00.

5.5 Backup Verification

The script was executed manually to verify its operation:

```
sudo /usr/local/bin/respaldo_home.sh
```

The /var/backups directory was then reviewed, and the generated file was confirmed to exist. This evidence validates backup creation; by itself, it does not constitute a restoration test or a complete integrity verification of the archive.

```

oscar@oscar-VirtualBox:/$ ls -lh /var/backups
total 3.1M
-rw-r--r-- 1 root root 50K May 6 00:00 alternatives.tar.0
-rw-r--r-- 1 root root 90K May 6 22:15 apt.extended_states.0
-rw-r--r-- 1 root root 0 May 7 00:00 dpkg.arch.0
-rw-r--r-- 1 root root 32 May 6 00:00 dpkg.arch.1.gz
-rw-r--r-- 1 root root 1.5K May 5 17:25 dpkg.diversions.0
-rw-r--r-- 1 root root 323 May 5 17:25 dpkg.diversions.1.gz
-rw-r--r-- 1 root root 313 Apr 22 19:20 dpkg.statoverride.0
-rw-r--r-- 1 root root 205 Apr 22 19:20 dpkg.statoverride.1.gz
-rw-r--r-- 1 root root 1.5M May 6 22:15 dpkg.status.0
-rw-r--r-- 1 root root 354K May 5 17:26 dpkg.status.1.gz
-rw-r--r-- 1 root root 1.1M May 6 02:00 respaldo_home_2026-05-06.tar.gz

```

Evidence 10. File respaldo_home_2026-05-06.tar.gz generated in /var/backups.

6. Conclusion

Verified result	Evidence/configuration
Operational Linux environment	Ubuntu 26.04 LTS on Oracle VirtualBox
Users and groups	jdoe, asmith, bwhite; groups admin and desarrolladores
Shared permissions	770 + SGID on /home/proyectos and /home/admin
Automated backup	respaldo_home.sh + daily cron at 02:00
Generated file	respaldo_home_2026-05-06.tar.gz in /var/backups

During this project, an Ubuntu 26.04 LTS environment was configured in Oracle VirtualBox and essential Linux administration tasks were practiced, including system updates, user and group management, directory permissions, and backup automation.

Configuring groups and permissions helped demonstrate how access to shared resources can be controlled and how SGID can preserve group ownership for newly created files. Creating the backup script and scheduling it with cron also demonstrated how recurring administrative tasks can be automated using standard Linux tools.

The verification steps confirmed the expected behavior of each component within the scope of the exercise. The process also reinforced the importance of documenting configurations, not only as evidence of completed work but also as support for future maintenance, auditing, troubleshooting, and environment improvement tasks.